

James Lawrence

Graduate Software Engineer – Keele University, Class of 2025

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Professional Summary

Graduate Software Engineer with 7+ years of programming experience, including production Java development for systems supporting 4M+ monthly active users. specializing in backend systems, large-scale multiplayer services, and full-stack web development.

Experienced in Java, TypeScript, and Python, with production systems supporting millions of users. Seeking a graduate/junior software engineering role with strong technical ownership and growth.

Key Skills

- **Languages:** Java, TypeScript, JavaScript, Python, C#, C++, Swift
- **Frameworks & Platforms:** React, Material UI, Django, Bukkit/Spigot, Shopify (Liquid)
- **Infrastructure & Tools:** Git, GitHub, VS Code, GitHub Copilot, Gemini, REST APIs, CI/CD fundamentals, Jira, Linux
- **Core Strengths:** Scalable backend systems, concurrency, OOP, performance tuning, production debugging, Agile/Scrum

Education

Keele University

(2022 – 2025)

First Class Honours, Bsc Computer Science (Software Engineering)

Wolverhampton Royal School

(2020 – 2022)

A Levels in Computer Science, Sociology and Physics

Previous Experience

Your Coop (Feb 2026 – May 2026)

Customer Service Assistant

- Maintained high operational standards in a fast-paced retail environment, managing POS transactions, inventory replenishment, and complex customer queries
- Demonstrated strong reliability and team collaboration skills to ensure business continuity and compliance with health and safety regulations.

Cue And Dart World (Aug 2025 - Jan 2026)

IT & Full Stack Software Engineer

- Built and maintained a full Shopify e-commerce platform, including **custom Liquid development** and third-party integrations allowing consumers to have a smooth shopping experience.
- Developed middleware to synchronize **POS**, inventory, and online systems, reducing manual workload and data errors
- Managed on-site IT infrastructure (**networking, routers, POS, Wi-Fi**) across multiple retail locations ensuring that multiple locations communicate effectively
- Delivered ongoing technical support and advised leadership on cost-effective technical improvements

CubeCraft Games (Ziax LTD) *(Oct 2023 - Jan 2024)*

Java Software Developer

- Engineered and Refined High-Performance Minigames: Developed Java-based game logic and mechanics for a global Minecraft network, ensuring seamless gameplay across diverse player environments.
- Architected Cross-Platform Compatibility: Leveraged **CubeTap** (a specialized Paper Spigot fork) to facilitate cross-play between Bedrock and Java editions, enabling a unified backend infrastructure for a multi-platform user base.
- Optimized Systems for Mass Scale: Maintained and scaled server-side systems supporting **42,000 peak concurrent players and 4 million monthly active users (MAU)**, consistently hitting performance benchmarks.
- Resolved Critical Production Bottlenecks: Diagnosed and neutralized high-priority performance leaks and gameplay bugs on live production servers, directly improving server tick rates and player retention.

Furcation LTD *(Jul 2022 - Jan 2025)*

Software Developer

- Developed Full-Stack Registration Architecture: Collaborated within a four-person agile team to build a robust registration system using TypeScript, React (Material UI), and Django, streamlining the onboarding process for over **4,000 total users**.
- Optimized Database Performance: Designed and managed a **PostgreSQL** database schema to handle high-concurrency traffic, ensuring data integrity and sub-second response times during **peak loads of 1,000+ simultaneous users**.
- Delivered Secure API Integrations: Engineered secure **RESTful APIs** to bridge the frontend and backend, implementing best practices for authentication and data validation to protect sensitive user information.
- Iterative Feature Deployment: Led the continuous refinement of UI/UX components and backend logic, utilizing user feedback to iterate on features and improve the overall platform stability.

Selected Projects

KeeleMC - Distributed Game Server Platform (Java & PostgreSQL)

- Engineered a distributed Java network architecture that synchronized player data across **4 server instances** using a **custom Velocity proxy plugin**.
- Architected a modular core via the **PaperMC API**, reducing plugin overhead by 60% through optimized event listeners and a centralized command framework.
- Developed a **PostgreSQL-backed** minigame framework (UniWars) that automated 100% of the match lifecycle and managed real-time leaderboard tracking for unique players.

Web Integration & Backend - Full-Stack Infrastructure (JavaScript & Node.js)

- Built a centralized **Node.js REST API** to bridge game and web platforms, enabling real-time **synchronization of statistics** for users.
- Implemented RESTful endpoints and secure database queries to handle high-frequency data requests from both the web frontend and game clients.

FxRAT - Remote Administration & Security Framework (C#)

- Architected a multi-tier remote administration tool using a Client-Server model with a shared library for streamlined data serialization.
- Implemented **low-level system hooks** and secure socket communication to manage remote processes and file system operations.
- Developed a modular plugin system allowing for the dynamic execution of security tasks and system diagnostics on remote endpoints.

Game-Concept - Physics-Driven Desktop Application (C++)

- Developed a high-fidelity **3D simulation** using the Open Dynamics Engine (**ODE**), maintaining a consistent 60 FPS during complex collisions involving 16 rigid bodies.
- Optimized a **custom C++ rendering pipeline** within OpenFrameworks, reducing memory consumption by 30% through efficient object state management.
- Refined **CPU cycle allocation** to prioritize physics calculations, ensuring stable performance during high-intensity user input and real-time environment updates.

References

Zed Spencer-Milnes

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